

KAWEEN HASARANGA

Materials Science & Engineering Undergraduate

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PROFILE

Materials Science and Engineering undergraduate at the University of Moratuwa with a strong interest in polymer materials, natural rubber, polymer processing and materials characterization. Seeking an industrial internship to gain practical experience in rubber technology, manufacturing processes and materials testing. Interested in applying engineering knowledge to improve material performance, product quality and manufacturing efficiency.

EDUCATION

B.Sc. (Hons) in Engineering — Materials Science and Engineering 2024 – Present

University of Moratuwa, Moratuwa, Sri Lanka | CGPA: 3.10 / 4.00

G.C.E. Advanced Level — Physical Science Stream 2022

G/Kularathna College, Ambalangoda, Sri Lanka | 3 A Passes | Z-Score: 2.2651

TECHNICAL SKILLS

Polymer & Latex: Natural Rubber Latex, Latex Compounding, Polymer Processing, Rheology

Materials Characterization: Mechanical Testing, XRD, SEM, Optical Microscopy, Materials Analysis

Manufacturing & Quality: Materials Testing, Quality Control Fundamentals

Engineering Design: AutoCAD, Solid Edge, Engineering Drawing, 3D Modelling

Data Analysis: Origin, ImageJ, Microsoft Excel, Experimental Data Analysis

Programming: Python

Microsoft Office: Word, Excel, PowerPoint

ACADEMIC PROJECTS

Automatic Chicken Coop Door System 2025

Engineering Design Project — Finalist, IESL

- Developed an automated door system for chicken coops to improve poultry management and reduce manual operation.
- Investigated automatic opening/closing, time-based operation, environmental conditions, user safety and reliable mechanical movement.
- Integrated mechanical components, sensors, control electronics and an actuator mechanism, followed by testing and evaluation of system performance.

Ammonia Spoilage in Meat: Kinetic Analysis 2026

Materials / Kinetic Project

- Investigated ammonia formation during meat spoilage as a potential indicator of deterioration and food quality.
- Studied the relationship between ammonia concentration, spoilage time and environmental conditions, focusing on reaction kinetics of ammonia generation.
- Applied kinetic models and experimental data analysis to evaluate ammonia monitoring as a spoilage-detection method.

Gearbox Design and Power Transmission Analysis 2026

Mechanics of Machines / Engineering Design Project

- Designed and analyzed a gearbox system for power transmission, torque control, and speed modification.
- Evaluated gear selection, gear ratios, torque requirements, and drum speed using engineering calculations.
- Developed suitable gear arrangements to meet mechanical system performance requirements.

Quantum Dot Optical Emission for Cancer Detection 2026

Nanomaterials / Electronic Materials Project

- Investigated quantum dots as optical sensing materials for potential cancer-detection applications, based on size-dependent optical properties.
- Studied the relationship between quantum-dot size, electronic structure, energy levels and photoluminescence emission for biomedical sensing suitability.
- Analyzed changes in optical emission as a detection mechanism and evaluated related material properties and operating principles.

SOFT SKILLS

Teamwork, Problem Solving, Technical Communication, Presentation Skills

LANGUAGES

Sinhala, English, Tamil (Basic Reading and Speaking)

ACADEMIC INTERESTS

Polymer Materials, Natural Rubber, Rubber Technology, Polymer Processing, Materials Characterization, Materials Testing, Quality Control, Manufacturing Processes, and Research & Development.

REFERENCES

Dr. A.M.P.B. Samarasekara - Senior Lecturer, Dept. of Materials Science and Engineering, University of Moratuwa
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Prof. D. Attygala - Senior Lecturer, Dept. of Materials Science and Engineering, University of Moratuwa
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